Experiment-6

Aim: Create Product Catalog Application using ASP.NET MVC Architecture (Controllers, Models, Views, Routing)

Code:

ProductControllar.cs

```
using prac_6.Models;
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Linq;
using System.Web;
using System.Web.Mvc;

namespace prac_6.Controllers
{
    public class ProductController : Controller
    {
        // GET: Product
        public ActionResult Index()
        {
            List<Product> P = new List<Product>();
            P = GetProducts();
            return View(P);
        }
        public ActionResult Details(int id)
        {
            List<Product> PALL = GetProducts();
            Product P1 = PALL.FirstOrDefault(p => p.ProductId ==
id);
            return View(P1);
        }
        private List<Product> GetProducts()
        {
            return new List<Product> {
            new Product
            {
                ProductId = 1,
                ProductName = "mobile",
                Description = "samsung",
                Price = 5000,
                Category="electronics"
            },
            new Product
            {
                ProductId = 2,
                ProductName = "mobile",
                Description = "samsung",
                Price = 5000,
                Category = "electronics"
            },
            new Product
```

```
                {
                    ProductId = 3,
                    ProductName = "mobile",
                    Description = "samsung",
                    Price = 5000,
                    Category = "electronics"
                },
                };

            }
        }
    }
```

Product.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;

namespace prac_6.Models
{
    public class Product
    {
        public int ProductId {  get; set; }
        public string ProductName { get; set; }
        public string Description { get; set; }
        public double Price { get; set; }
        public string Category { get; set; }
    }
}
```

Details.cshtml

```
@model prac_6.Models.Product
@{
    ViewBag.Title = "Details";
}

@{
    Layout = null;
}
<h2>Product detail</h2>
@if (Model != null)
{
    <p>Product Id:@Model.ProductId</p>
    <p>Product Name:@Model.ProductName</p>
    <p>Destription:@Model.Description</p>
    <p>Price:@Model.Price</p>
    <p>Category:@Model.Category</p>

}
else
                {
                    <p>Product not found</p>
                }
```

```
<br/>
@Html.ActionLink("Bck to product catelogue", "Index", "Product");


Index.cshtml
@model List<prac_6.Models.Product>
    @{
        ViewBag.Title = "Index";
    }

@{
    Layout = null;
}

<h2>Product Catalogue</h2>
<table border="1">
    <tr>
        <th>Product Id</th>
        <th>Product Name</th>
        <th>Description</th>
        <th>Price</th>
        <th>Category</th>
    </tr>
    @foreach(var product in Model)
            {
                <tr>
                    <td>@product.ProductId</td>
                    <td>@product.ProductName</td>
                    <td>@product.Description</td>
                    <td>@product.Price</td>
                    <td>@product.Category</td>
                    <td>@Html.ActionLink("view
details","Details","Product"), new {id=product.ProductId}, null</td>
                </tr>
            }
</table>

RouteConfig.cs
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Mvc;
using System.Web.Routing;

namespace prac_6
{
    public class RouteConfig
    {
        public static void RegisterRoutes(RouteCollection routes)
        {
            routes.IgnoreRoute("{resource}.axd/{*pathInfo}");

            routes.MapRoute(
```

```
                name: "Default",
                url: "{controller}/{action}/{id}",
                defaults: new { controller = "Product", action =
"Index", id = UrlParameter.Optional }
            );
        }
    }
}
```

Global.asax.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Mvc;
using System.Web.Optimization;
using System.Web.Routing;

namespace prac_6
{
    public class MvcApplication : System.Web.HttpApplication
    {
        protected void Application_Start()
        {
            AreaRegistration.RegisterAllAreas();

            RouteConfig.RegisterRoutes(RouteTable.Routes);

        }
    }
}
```

Output – Screenshot:

Product Catalogue

Vrushti Satodiya (92400103139-batch-B)

Product Id	Product Name	Description	Price	Category	
1	Samsung S23 Ultra	Galaxy S23 Ultra 256GB, 12GB RAM	87000	Electronics	view details
2	MacBook	Apple MacBook Pro 16-inch, M2 Pro chip	127000	Electronics	view details
3	Tablet	Lenovo Tab	5000	Electronics	view details
4	Mobile Cover	Mobile cover for iphone 17 pro max	1700	Mobile Accessories	view details
5	Laptop Bag	Laptop bag for 15.6 inch	548	Laptop Accessories	view details
6	Earphone	MIVI wireless earphone	799	Earphones	view details
7	Tablet	Lenovo	8000	Electronics	view details
8	boAt Karaoke	boAt PartyPal Karaoke, 90w output	6999	Home Audio	view details
9	PS5	Sony PS5 Console Digital	64990	Gaming	view details
10	Vara Art Showpiece	Vara Art Handcrafted Mental Rajasthani showpiece	667	Home Decore	view details

Product detail

Product Id:5

Product Name:Laptop Bag

Destription:Laptop bag for 15.6 inch

Price:548

Category:Laptop Accessories

Back to product catalogue: